Design Implementation of A Hybrid VLP/PLC-Based Indoor Tracking System for Smart Hospitals

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***Abstract*—Modern healthcare systems will necessitate smart hospital operations with system-wide wireless communications and positioning capabilities, which cannot be entirely facilitated by existing radio-frequency (RF) technologies due to interference concerns. We present design and implementation of a novel hybrid wireless tracking system utilizing visible light communication (VLC) and positioning (VLP) technologies, aided by powerline communication (PLC), for smart hospital operations. Utilizing the existing LED lighting infrastructure, this VLC/VLP/PLC positioning system consists of host optical transceivers embedded in light-emitting diode (LED) bulbs and user-end photodetector (PD) optical tags to form a wireless tracking network connecting all people and equipment throughout all hospital buildings. The new indoor optical tracking system was validated via simulation and experiments. Smart hospital operations will lead to more affordable healthcare with improved quality in the near future.**

Keywords—Indoor positioning system, visible light positioning, VLP, visible light communication, VLC, powerline communication, PLC, received signal strength, LED

I. Introduction

Wireless communication and positioning technologies have altered our civilization as one of the fastest-growing application sectors. Unlicensed optical wireless communication has recently been considered as an alternative to existing radio frequency (RF) communications and positioning methods, especially in hospital settings where RF wireless is forbidden in critical wards since strong RF interferences may be life-threatening due to interactions with medical devices [1]. To address this critical need for smart hospital operations in an indoor environment, we proposed and designed an LED-based VLC/VLP technology platform to realize a low-cost, low-complexity indoor tracking system for smart hospitals. The new VLC/VLP smart hospital system utilizes the existing LED lighting infrastructure in a hospital to realize an energy-efficient visible light tracking system, which enables real-time and precise asset and person tracking and, hence, will dramatically reduce healthcare costs by making hospital operations more efficient and greener. This LED VLC/VLP system allows for big data collection, development of user-hospital-building interaction models, and implementation of artificial intelligence (AI) algorithms for future smart hospital operations [2-5]. In this work, visible light, infrared (IR) light and powerline communications are combined to provide LED-based VLP functions within a building and support infrastructure-wide data communications for large-scale smart hospital operations.

II. System Design

The new VLC/VLP system utilizes two unidirectional lines of communication to achieve bi-directional communications between tracked users (e.g., actors, medical devices) and light fixtures housing LED bulbs. The download line of communication uses on-off keying to transmit data through visible light, while the upload uses infrared signals to avoid interference with human eyes. In the VLC/VLP system, each fixture consists of a four-by-four array of commercial LEDs (shown in Fig.1), an infrared receiver, an Arduino Uno microcontroller with an ethernet shield, an amplifier circuit used to drive the LEDs with 28V, and a connection to the powerline through a Powerline 500 AV Nano Adapter. Fig. 2 depicts the block diagrams of the tracking fixture units and the device used to track a user. The user device contains two microcontrollers,

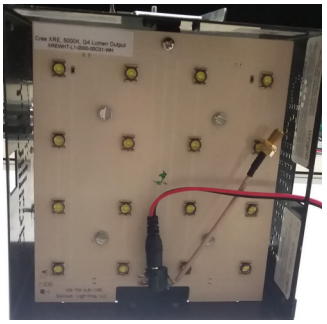


Fig. 1. The underside of a fixture unit.

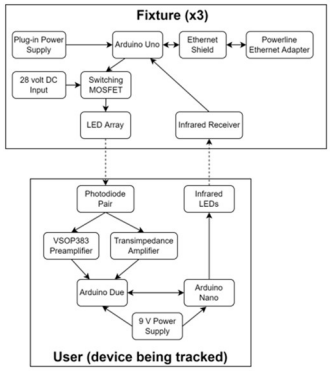


Fig. 2. Block diagrams for the tracking fixtures and the tracked device.

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an Arduino Due and an Arduino Nano, which separately handle receiving visible light signals and transmitting infrared light signals. The output of the VSOP383 preamplifier is a digital signal which provides the binary data transmitted by a fixture. The transimpedance amplifier (TIA) produces an analog signal, which is used to determine the received signal strength. The Arduino Nano then transmits data back to the fixtures using infrared LEDs. The data is encoded using the NEC protocol with a 38KHz carrier frequency.

III. VLC Tracking Theory and Implementation

To encode the ID of a fixture into its visible light signals, each LED fixture's microcontroller can alternate between outputting a constant voltage and outputting a 38KHz square wave on the pin. When sending a zero ("0") bit, the microcontroller outputs the constant voltage. When sending the one ("1") bit, the microcontroller outputs the 38KHz square wave. Fig. 3 breaks down the sequence and timing for sending one or more bytes of data. The start bit for the first byte transmitted needs to last 2000 microseconds to provide the user with sufficient time to collect the samples used to determine the received signal strength. All other bits last for 300 microseconds. The total time required to send the first byte in a sequence is 4700 microseconds, and for consecutive bytes it is 3000 microseconds.

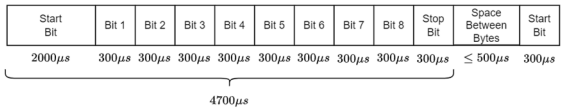


Fig. 3. Sequencing and timing of bits transmitted by a tracking fixture when multiple bytes are transmitted. The transmission ends after the first stop bit.

The signal from the photodiode is also amplified as an analog signal by the TIA. An analog to digital converter (ADC) built into the microcontroller samples the output from the TIA during the start bit to measure the signal's average peak-to-peak (p-p) amplitude (A_{p-p}) which is used to estimate the distance between the user and the transmitting fixture. An optical channel model used in [6] could be applied to relate the RSS to the distance between the fixture and the user. With a reasonable approximation and assuming the Lambertian order of the LEDs to be one, the DC gain can be simplified with component of the distance along the ground plane (d_{xy}) as the only value that is variable:

$$H(d) = \frac{K(d_z)^2}{\pi\left[(d_{xy})^2+(d_z)^2\right]^2} \quad (1)$$

where K is a constant and d_z is the vertical component of the distance. To eliminate the error of approximation, calibration is performed under conditions similar to those under which the system needs to operate in a real world. This is done by measuring A_{p-p} while varying the distance between the fixture and the photodiode. This method assumes that the photodiode will maintain a fixed height relative to the fixtures at all times.

For the calibration setup, the user side were fixed to a slider that could move along one axis and a motor was used to automatically pull the slider at a low speed in order to vary the distance between the photodiode and the fixture. Measurements were taken for every 6 mm that the photodiode moved. At each measurement point, the Arduino Due took 1000 samples of the signal from the TIA using its ADC to compute A_{p-p}. A_{p-p} is expressed in terms of LSBs of the ADC output.

Fig. 4 shows a set of calibration data collected for d_{xy} ranging from 0 m to 2.864 m. Each data point is indicated with a blue dot and indicates an average amplitude measurement obtained at a specific distance from the fixture. Superimposed onto the data is a piecewise linear function consisting of 13 line segments, shown in black. Each line segment covers a different interval of the x-axis with the equation for each line segment being of the linear form

$$d_{xy} = c_1 \cdot A_{p-p} + c_2 \quad (2)$$

where c_1 and c_2 are constants determined using the MATLAB Curve Fitting Toolbox. Together, these equations are used to model the relationship between signal amplitude (A_{p-p}) and distance (d_{xy}), so that the distance can be quickly computed on the Arduino Due.

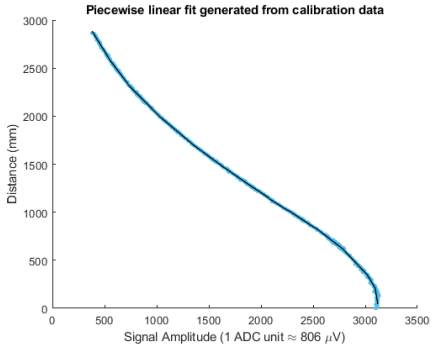


Fig. 4. Scatter plot of the full set of calibration data collected for a single fixture, with the piecewise linear function used to fit the data overlaid in black.

We have evaluated our VLC/VLP system in terms of the accuracy of distance measurements between the user and an individual fixture, as well as the rate at which new location estimates are produced. We found that distance measurements are generally accurate to within 10cm, although the accuracy is degraded when the user is within 50cm of the transmitting fixture. When the system is running continuously, a new location estimation consisting of a distance for each of the three fixtures is added to the CSV file at least once every 1.7 seconds.

IV. GUI Subsystem

The GUI mapping subsystem implements the front end of the VLP/PLC tracking system, providing three main features: data input, data processing, and data output. These three features are implemented and interface with the GUI using Python. The GUI subsystem receives data from visible light tracking through the utilization of powerline communication. The data is transmitted to the computer system through UART and logged into a CSV file, which is readily accessed via Python. The GUI subsystem processes user data using triangulation of the aforementioned distance data by utilizing Python Descartes and Shapely libraries. Then GUI subsystem implements a window including all relevant light fixture data and plotting of

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triangulated data using the Tkinter library. The GUI automatically reads the data from the CSV file and updates the plot with the user positions in rectangular coordinates. Fig. 5 shows the region of intersection with the three LED intensity rings. The tracking results will be further used to construct a tracking map showing all actors and equipment monitored in a hospital, overlaying the given campus and building layout to facilitate smart tracking and operations.

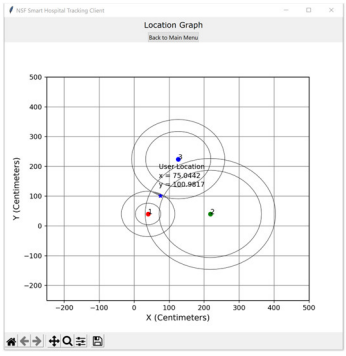


Fig. 5. GUI shows user position as (75.0442,100.9817) with the distance of 562.91mm, 1663.95mm, and 1131.54mm from LED 1, 2 and 3.

V. Conclusion

We present the design and implementation of a novel hybrid visible light and powerline communication enabled indoor positioning system for smart hospital applications. The new indoor all-optical tracking system includes host LED lamps, user-end optical tags, triangular LED positioning algorithms and a PLC interface for decimeter-level tracking accuracy. The new VLC/VLP system has been experimentally validated at board level. Integrated circuits (ICs) for this VLC/VLP tracking system are designed and will be fabricated in a 45nm SOI CMOS technology. This VLC monitoring system can potentially revolutionize hospital operations in the future, leading to cost-effective and greener healthcare.

Acknowledgment

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